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Imperial College London




**FACULTY OF ENGINEERING**

**DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING**

# Smart Grid Technology

COURSEWORK I REPORT

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SUBMITED TO
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## CERTIFICATE OF ORIGINALITY

This is to certify that I am responsible for the work submitted in this project, that the original work is my own except as specified in the references, and that the original work contained herein have not been taken or done by unspecified sources or person.

_____

Signature:

Motaz M. Ayiad, _____

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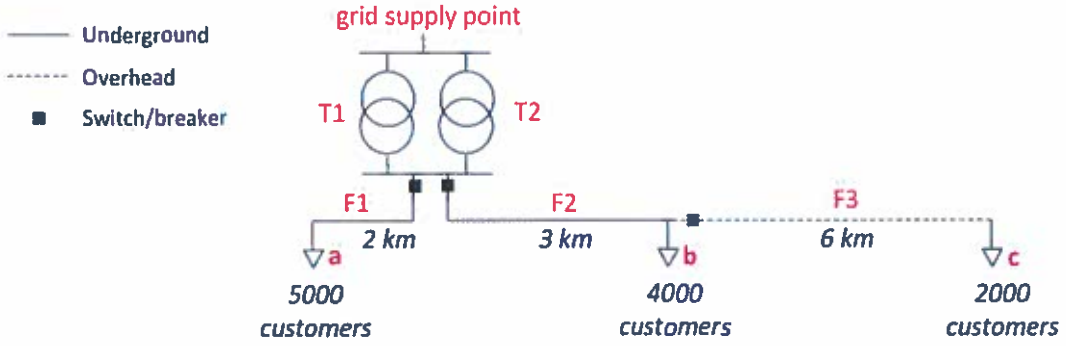
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# CHAPTER 1

# COURSEWORK

## 1.1 INTRODUCTION

In this report, the **reliability** of a simple distribution system will be analyzed using three approaches: analytical, snapshot Monte Carlo simulations and time-sequential Monte Carlo simulations. The distribution network is depicted below. The diagram lists the number of customers (residential households) at each load point and the lengths and types of connection. The transformers have a thermal rating of 8MVA each and it may be assumed that the transformer will trip when overloaded. The study is carried out under ideal circumstances [1].




However, the loads are described by the average daily load profile shown in Figure 1-1 [2].

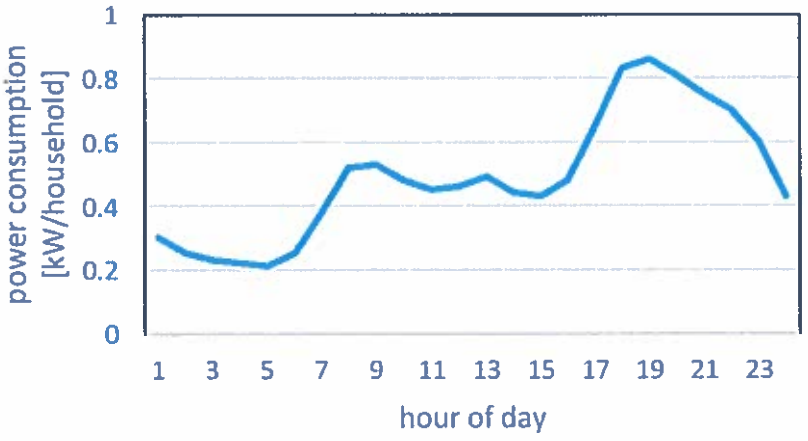



**Figure 1-1- Average daily load profile**

The components failure and repair parameters are summarized below, Table 1-1.

**TABLE 1-1**, Failure and Repair Parameters.

| Component | Fault process | Repair process |
|---|---|---|
| Transformer (T1,T2) | MTTF = 100 years | MTTR = 10 days |
| Overhead cable (F3) | Failure rate = 0.1/km/year | MTTR = 10 hours |
| Underground cable (F1,F2) | Failure rate = 0.05/km/year | MTTR = 2 days |

## 1.2 SOLUTION

In this section, three different approaches will be carried in order to identify the reliability of the whole system (network), but the first stage for all methods is to find the availability of all the components.

By modifying Table-1 and taking the following definitions, a simple calculations can be written.

Mean time to failure (MTTF): the average operational time, MTTF $= \frac{1}{FailureRate \times N}\ years.$

Mean time to repair (MTTR): the average repair/replacement time.

Mean time between failures (MTBF): average time for a complete failure-repair cycle (MTBF = MTTF + MTTR).

Failure rate ($\lambda$): $\lambda = \frac{1}{MTTF}\ 1/unit/years$ and Repair rate ($\mu$): $\mu = \frac{1}{MTTR}\ 1/unit/years$

Availability: proportion of time spent in the up state

$$P(up) = \frac{MTTF}{MTBF} = \frac{MTTF}{MTTF + MTTR} = \frac{\mu}{\lambda + \mu} \quad [1]$$

F1, F2 and F3 are feeders where their MTTF depend on the cables length, the length of the feeders are; 2, 3 and 6 Km respectively, from this the calculation of MTTF for the feeder can be carried easily as follow,

$$MTTF_{F1}\ \frac{1}{0.05 \times 2} = 10\ years = 3650\ days$$

$$MTTF_{F2}\ \frac{1}{0.05 \times 3} = \frac{20}{3}\ years = \frac{7300}{3}\ days$$

$$MTTF_{F3}\ \frac{1}{0.1 \times 6} = \frac{5}{3}\ years = \frac{1825}{3}\ days$$

# Summary of Comments on Blank Project Management Plan Template

## Page: 9

Number: 1 Author: Simon Tindemans Date: Indeterminate

Simon Tindemans: Note that this last step assumes that both failure and repair steps have exponential distributions. That has not been assumed in the exercise: the repair process can be anything with a mean repair time given by MTTR!

The modified table can be shown in Table 1-2, which can be used to calculate the availably, Note: all units must be unified (i.e. in days)

**TABLE 1-2-** Ready-To-Use Failure and Repair Parameters.

| Component | Fault process | Repair process |
|---|---|---|
| Transformer (T1,T2) | MTTF = 36500 days | MTTR = 10 days |
| Overhead cable (F3) | MTTF = 1825/3 days | MTTR = 10/24 day |
| Underground cable (F1) | MTTF = 3650 days | MTTR = 2 days |
| Underground cable (F2) | MTTF = 7300/3 days | MTTR = 2 days |

Availability can be found now, Table 1-3:

$$Av_{T1\&T2} = \frac{36500}{36500 + 10} = 0.999726 \qquad Av_{F1} = \frac{3650}{3650 + 2} = 0.999452$$

$$Av_{F2} = \frac{7300/3}{7300/3 + 2} = 0.999179 \qquad Av_{F3} = \frac{1825/3}{1825/3 + 10/24} = 0.999316$$

ok

**TABLE 1-3-** Components Availabilities.

| Component | Fault process | Repair process | Availability |
|---|---|---|---|
| Transformer (T1,T2) | MTTF = 36500 days | MTTR = 10 days | 0.999726 |
| Overhead cable (F3) | MTTF = 1825/3 days | MTTR = 10/24 day | 0.999316 |
| Underground cable (F1) | MTTF = 3650 days | MTTR = 2 days | 0.999452 |
| Underground cable (F2) | MTTF = 7300/3 days | MTTR = 2 days | 0.999179 |

### 1.2.1 ANALYTICAL APPROACH

In this sub-section, the analytical approach is used to find the reliability of the whole system, this method requires the components availabilities and the probability of the loads will exceed the rated transformer capacity. This method can also be used to find the loss of load probability (LOLP) and Customer minutes lost CML.

$$P(Load \times TotalCustomrs > 8000) = \frac{4\ hours}{24\ Hours}$$

$$P(Load \times TotalCustomrs < 8000) = \frac{20\ hours}{24\ Hours}$$

From the above, the probability of each feeder stay up can be calculated as follow,

$$P(F1) = P(Load > 8000).P(T1\ and\ T2).Av_{F1} + P(Load < 8000).P(T1\ or\ T2).Av_{F1}$$

$$P(F1) = \left(\frac{4}{24}\right).(0.999726 \times 0.999726).(0.999452) +$$

$$\left(\frac{20}{24}\right).(1-(1-0.999726)^2).(0.999452)$$

$$P(F1) = 0.999361$$

In the same way for feeder 2.

$$P(F2) = 0.999087$$

However, feeder 3 has some differences because it depends on the availability of feeder 2,

$$P(F3) = P(F2).Av_{F3} = 0.999087 \times 0.999316 = 0.998404$$

With the knowledge of all the probabilities, a table with the approach targets can be construct, as shown in Table 1-4.

**TABLE 1-4**- Final Analytical Approach Results.

| Feeders Customers | Probability to stay up | Customers stay up |
|---|---|---|
| F1 = 5000 | 0.999361 | 4996.81 |
| F2 = 4000 | 0.999087 | 3996.35 |
| F3 = 2000 | 0.998404 | 1996.81 |
| Total Customers = 11000 | | |
| System Availability | $\frac{Sum(Customers\ stay\ up)}{Total\ Customers} = 0.9990875$ | |
| LOLP | $1 - \text{System Availability} = 0.0009125$ | |
| CML | $LOLP \times 60 \times 24 \times 365 = 479.61\ minutes$ | |

# Page: 11

Number: 1 Author: Simon Tindemans Date: Indeterminate

Simon Tindemans: Some issues:
1) no explicit computation of unavailability (as request ed)
2) no recognition of the fact that remote disconnections can drop the effective system load below 8MW. For example, a fault on F2 at peak load results in a load <8MW, so that only a single transformer is required to supply load point a.

Number: 2 Author: Simon Tindemans Date: Indeterminate

Simon Tindemans: LOLP is fraction of time that any customer (anywhere in the system) is disconnected. With this definition of system availability, something else is computed here.

Note that you cannot compute LOLP from load point availabilities, because they don't tell you about dependencies between failures.

Number: 3 Author: Simon Tindemans Date: Indeterminate

Simon Tindemans: The CML has not been compared with that of real distribution networks.

### 1.2.2 SNAPSHOT MONTE CARLO APPROACH

In this sub-section, snapshot Monte Carlo simulation used to find the availability of the system (or LOLP) using MATLAB software, the codes are in appendix-A (original codes before modification are in [3]).

This method can be used to estimate the loss of load probability (LOLP), Customer minutes lost (CML) and the Expected energy not supplied (EENS).

The concept behind this method is to use the components availability to generate random states of these components then find the impacts of the states on the system which in the end can be used to estimate the objective values, but to achieve that the simulation must run for a long numbers of samples such as 100K samples. Hence more samples more accurate estimation.

In following figures (1-2 and 1-3) show the results – with errors- of one Snapshot Monte Carlo simulation test. However detailed comparisons between the three approaches is listed in section 1.3.

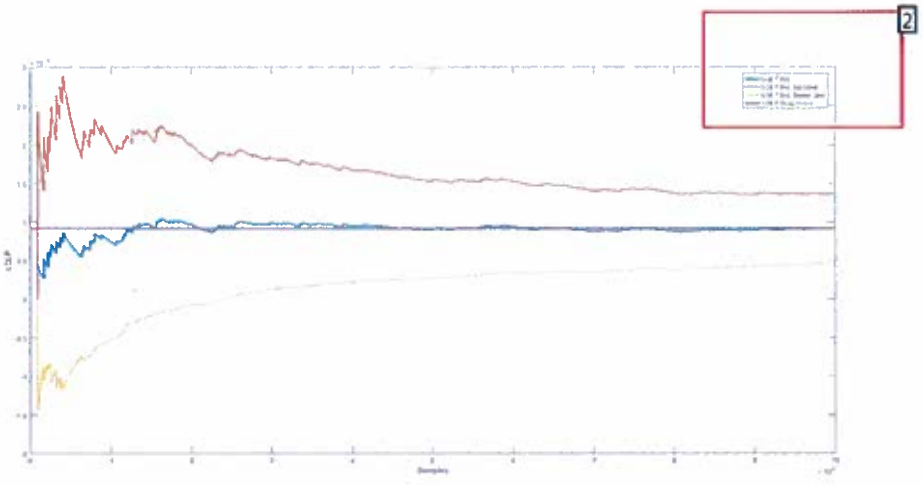


**Figure 1-2- Snapshot: LOLP estimate with up/down limits**

```
LOLP = 0.00092 ± 0.00007
CML = 481.64073 mins ± 36.38295
EENS = 46380 KWh ± 4358.13388
```

**Figure 1-3- Snapshot: LOLP, CML and EENS estimate**

# Page: 12

Number: 1 Author: Simon Tindemans Date: Indeterminate
Simon Tindemans: have you defined the state? (question 6)

Number: 2 Author: Simon Tindemans Date: Indeterminate
Simon Tindemans: explain what the lines are (I can guess, but the text should mention this)

Number: 3 Author: Simon Tindemans Date: Indeterminate
Simon Tindemans: Have you used the same code for this result as in the appendix? It seems to compute the error in MWh (1000x smaller)

### 1.2.3 TIME-SEQUENTIAL MONTE CARLO APPROACH

In this sub-section, time-sequential Monte Carlo simulation used to find the availability of the system (or LOLP) using MATLAB software, the codes are in appendix-A (original codes before modification [4]).

This method can be used to estimate the loss of load probability (LOLP), Customer minutes lost (CML), Customer Interruptions (CI) and the Expected energy not supplied (EENS).

The concept behind this method is to use the components availability to generate random states of these components then find the impacts of the states on the system over a period of time such as a 1000 year, and each year the states are changed in a small period of time called Delta-t and this process depends on Markov chain (two states with repair & failure rates) and this process yields to estimate the objectives, this simulation can run for a long period of time such as 1000 years with 1 Hour Delta-t. Hence more years and small Delta-t gives more accurate estimation.

Figure 1-4, shows the results – with errors- of one Time-Sequential Monte Carlo simulation test. However detailed comparisons between the three approaches is listed in section 1.3.

```
LOLP  = 0.00092 - [0.00080, 0.00105] (95% confidence)
CML   = 484.49455 - [418.45517, 550.53393] (95% confidence)
CI    = 40.47273 - [40.36660, 40.57885] (95% confidence)
EENS  = 45904.39000 - [39271.41643, 52537.36357] (95% confidence)
```

**Figure 1-4- Time-Sequential: LOLP, CML, CI and EENS estimate**

# Page: 13

Number: 1 Author: Simon Tindemans Date: Indeterminate

Simon Tindemans: This range is suspiciously narrow. When you see this, it is a signal to take another good look at the code: there is a factor 100 missing.

Figure 1-5, shows two state Markov chain of each component with the failure and repair rates, λ and μ.

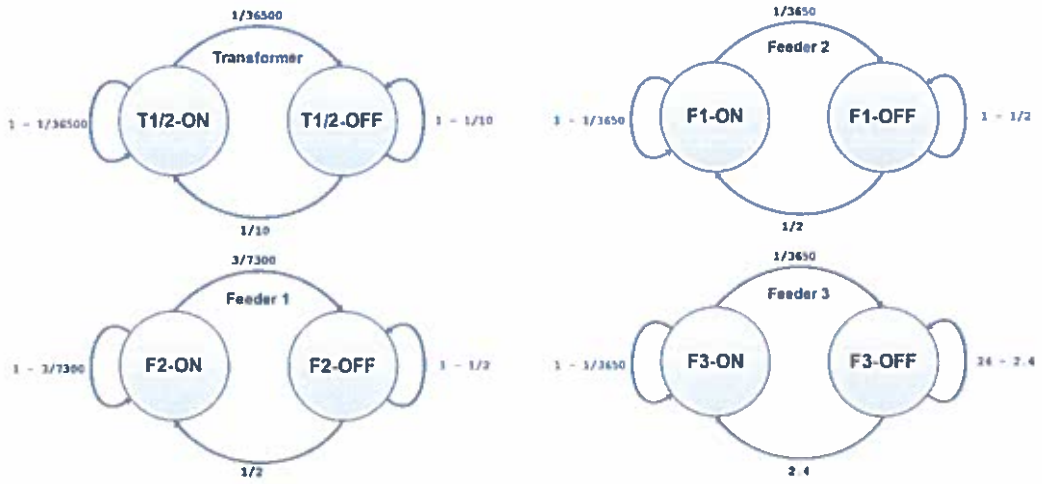



**Figure 1-5- Markov Chain of each component**

Figure 1-6, 1-7 and 1-8, show a histogram of each CML, CI and EENS estimated value, values in these plots are have one thing in common which is the number of occurrences is high at low values of CML, CI and EENS, however the high values in theses estimated values have a very low number of occurrences and the existing of such behavior depends of the availability of each component in the system.

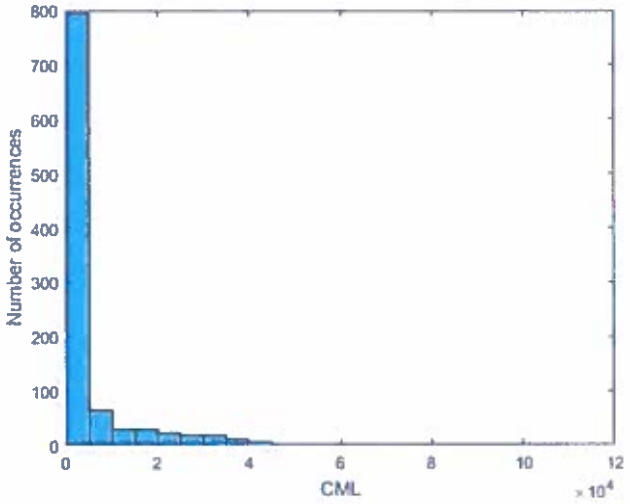



**Figure 1-6- CML Histogram representation**

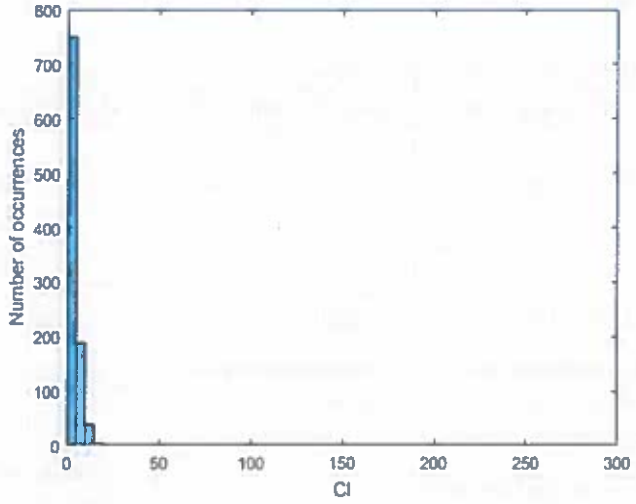



**Figure 1-7- CI Histogram representation**

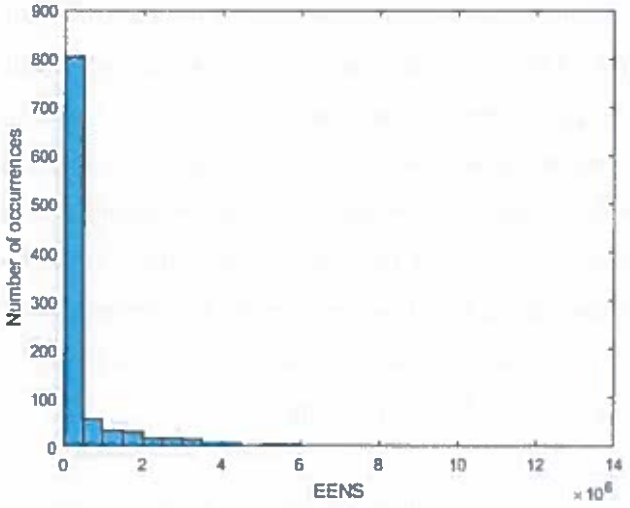



**Figure 1-8- EENS Histogram representation**

## 1.3 DISCUSSION

**TABLE 1-5-** Comparison between the Three Approaches.

| | Analytical approach | Snapshot approach | Time-Sequential approach |
|---|---|---|---|
| **LOLP** | 0.0009125 | 0.00092 | 0.00092 |
| **CML** | 479.61 *minutes* | 481.64 *minutes* | 484.494 *minutes* |
| **CI** | - | - | 40.47 |
| **EENS** | - | 46380 Kwh | 45904 Kwh |
| **Executing Time** | Approx. 1/2 hour | Approx. 10 sec. | Approx. 25 sec. |
| **Accuracy** | High accuracy, but depends on the calculation resolution | Less accuracy and depends on number of samples | Medium accuracy and depends on time and Delta-t |
| **Complexity** | Increasable and can be impossible for complex distribution network | Considered simple algorithm | Can be complete complex but still able to overcome the impossibility of analytical approaches |
| **Reliability** | Constant, always the same value | Different estimations around the true value | Different estimations around the true value |

Table 1-5, shows that an accurate solution such as the analytical approach got LOLP = 0.0009125 and $CML = 479.61\ minutes$, however the estimation of both values in the two simulation approaches got different values compared with the true value with less than 0.0000875 for LOLP, and less than 5 minutes for CML. Despite that, the analytical approach can't be used to find CI or EENS, while Snapshot and Time-Sequential can be used to estimate EENS[1], and CI can only be found using Time-Sequential method (not accurate true).

In general, Monte Carlo simulations are suitable for problems with multidimensional cases or when the terminal distribution cannot be computed, but with single dimension case, it is better and more accurate to use the analytical approach. ok

Despite that CML of 479.61 minutes per year seems low value, but it still not acceptable, especially when it compared with the CML of different countries in 2006, for instance the average U.S. customer loses power was 214 minutes per year. That compared to 70 in the United Kingdom, 53 in France, 29 in the Netherlands, 6 in Japan, and 2 minutes per year in Singapore [5].

[1] EENS should be around 47000 Kwh and CI should be around 41. This assumption comes after multiple runs and comparing the results with my classmates.

# Page: 16

Number: 1 Author: Simon Tindemans Date: Indeterminate
Simon Tindemans: nice comparison

Number: 2 Author: Simon Tindemans Date: Indeterminate
Simon Tindemans: Note that you should always include error estimates when discussing MC results. Without it, readers do not know how to interpret these numbers.

Number: 3 Author: Simon Tindemans Date: Indeterminate
Simon Tindemans: Comparisons require mention of confidence intervals.

1

The last thing to mention, Figure 1-6 and 1-7 show very high CML and CI at extreme years which is derived by the missing of management activities against the increase in load on the distribution system.

Distribution Network Operators or DNOs are now looking at using innovative means such as demand side management (DSM) to do so. This involves the DNO contracting with a network customer to constrain their demand at certain times. DNOs can enter into these types of arrangements to avoid the cost of network reinforcement [6].

2

DNOs activities can be in three different domains,

1- 'The rebound effect': it refers to behavioral or other systemic responses to the introduction of new technologies that increase the efficiency of resource use.
2- Working on the customer side of the meter: increase the efficiently through working with customers and installing infrastructure on the customer side of the meter.
3- Go to other sources of funding for energy demand reduction: this could work to raise the profile of other energy demand reduction schemes amongst customers on its network and facilitate participation in energy demand reduction schemes such as: ECO and The Green Deal.

However these options require information to decide which one is cost-benefits. For example, action on the first domain requires types of energy demand reduction measures that have a 'direct impact' on the network. Which is defined as something would have an observable and controllable impact on the distribution system. The second domain requires equalizing the incentives and moving to a totex approach, in addition to the costs of investing in such infrastructure (smartmeters and data concentrators…etc.). The last domain requires participating in different energy demand reduction schemes which looks the low cost with predict benefits compared with the first option where collecting measures could be costly and require much effort and the second option is very costly and time consuming at the beginning approach regardless it could be a very good option to control costumer's loads.

3

**Outline-Comment:** the information on this report could shock people whose come from conflict areas –like I -, conflict areas could have CML of 175200 minutes per year or 8 hours per day under best conditions.

# Page: 17

Number: 1 Author: Simon Tindemans Date: Indeterminate

Simon Tindemans: But what happens during these years? For example: transformer outage leading to recurrent outages (every peak).

Number: 2 Author: Simon Tindemans Date: Indeterminate

Simon Tindemans: I do not really see these as three different options: customer-side options can involve funding through the Green Deal and result in a 'rebound effect'. These are more like aspects to consider when proposing demand side solutions.

Regarding costs: what about benefits? How would you quantify the benefits of improved reliability?

Number: 3 Author: Simon Tindemans Date: Indeterminate

Simon Tindemans: Yes, that is good to put such numbers in perspective.

# APPENDECES

## APENDIX A: MATLAB-CODE

### D.1 State Sampling/Snapshot Monte Carlo

```
% Main Script
SAMPLE_SIZE = 100000;
% initialise empty arrays
lolp_SYS = zeros(1, SAMPLE_SIZE);
lolp_A = zeros(1, SAMPLE_SIZE);
lolp_B = zeros(1, SAMPLE_SIZE);
lolp_C = zeros(1, SAMPLE_SIZE);
Load = zeros(1, SAMPLE_SIZE);
for i=1:SAMPLE_SIZE
    % generate samples and compute three measurements each time
    state = Get_Sample();
    [lolp_SYS(i), lolp_A(i), lolp_B(i), lolp_C(i)] = Get_Disc(state);
    Load(i) = state(6);
end
    EENS_values =  (lolp_A*5 + lolp_B*4  + lolp_B.*lolp_C*2).*(Load/11);

% compute and display LOLP, CML and EEND values
lolp_estimate = (mean(lolp_A)*5 + mean(lolp_B)*4 +  mean(lolp_C)*2)/11;
lolp_se = (std(lolp_A*5 + lolp_B*4 + lolp_C*2)/(11*sqrt(SAMPLE_SIZE)));
CML_estimate = (mean(lolp_A)*5 + mean(lolp_B)*4 +  mean(lolp_C)*2)*(24*60*365/11);
CML_se = (std(lolp_A*5 + lolp_B*4 + lolp_C*2)/(11*sqrt(SAMPLE_SIZE)))*(24*60*365);
EENS_shed_estimate = mean(EENS_values)*10000;
EENS_shed_se = std(EENS_values*10)/(sqrt(SAMPLE_SIZE));

fprintf('\n LOLP = %.5f ± %.5f\n',lolp_estimate, lolp_se);
fprintf(' CML = %.5f mins ± %.5f\n',CML_estimate, CML_se);
fprintf(' EENS = %.0f KWh ± %.5f\n\n',EENS_shed_estimate, EENS_shed_se*1000);
% for visualising the convergence, create an array of partial means and
% partial standard errors. Use only one in 10 samples to speed things up
partial_means = zeros(1, SAMPLE_SIZE/10);
partial_ses = zeros(1, SAMPLE_SIZE/10);
for i=1:SAMPLE_SIZE/10
    % compute partial means and partial standard errors
    partial_means(i) = (mean(lolp_A(1:10*i))*5 + mean(lolp_B(1:10*i))*4 +
mean(lolp_C(1:10*i))*2)/11;
    partial_ses(i) = std(lolp_A(1:10*i) + lolp_B(1:10*i) +
lolp_C(1:10*i))/sqrt(10*i);
end
% plot the true value, the MC estimate and the 95% confidence bands
plot(1:10:SAMPLE_SIZE, partial_means,'LineWidth',2);
hold on;
plot(1:10:SAMPLE_SIZE, partial_means + 1.96 * partial_ses,'LineWidth',2);
plot(1:10:SAMPLE_SIZE, partial_means - 1.96 * partial_ses,'LineWidth',2);
plot([1 SAMPLE_SIZE] , [0.000915 0.000915],'LineWidth',2);
xlabel('Samples');
ylabel('LOLP');
legend('LOLP Est.','LOLP Est. Up Limit', 'LOLP Est. Down Limit','LOLP True Value');
```

**Appendix Figure 1- Snapshot: Main Script – `RUN.m`**

```
function Load = Load_Profile( Hr )
Cust = 11000;
LD_Pf = [0.30 0.25 0.23 0.22 0.21 0.25 0.38 0.52 0.53 0.48 0.45 0.46 0.49 0.44 0.43
0.48 0.65 0.83 0.86 0.81 0.75 0.70 0.60 0.43];
MW_LD_Pf = Cust*LD_Pf;
Load = MW_LD_Pf(Hr);
End
```

**Appendix Figure 2- Snapshot: Load Profile Function, `Load_Profile.m`**

ok

Number: 1 Author: Simon Tindemans Date: Indeterminate
Simon Tindemans: It seems the EENS is given in kWh and its error in MWh?

```
function [ sampled_state ] = Get_Sample()
T1 = rand < 0.99972;
T2 = rand < 0.99972;
F1 = rand < 0.999452;
F2 = rand < 0.999178;
F3 = rand < 0.999315;
load = Load_Profile(floor(24*rand())+1);
sampled_state = [T1 T2 F1 F2 F3 load];
end
```

ok

**Appendix Figure 3- Snapshot: Get Samples Function, `Get_Sample.m`**

```
function [ C_Sys, C_a, C_b, C_c] = Get_Disc( state )
    C_Sys = 0; C_a = 0; C_b = 0; C_c = 0;

if (state(6) > 8000)
    if not((state(1) && state(2)))        % T1 and T2 Off
        C_Sys = 1; C_a = 1; C_b = 1; C_c = 1;
    elseif not(state(3) || (state(4))) % A&B Off
        C_Sys = 1; C_a = 1; C_b = 1; C_c = 1;
    else
        if state(3)
            C_a = 0;
        else
            C_a = 1; C_Sys = 1;
        end
        if state(4)
            C_b = 0;
            if state(5)
                C_c = 0;
            else
                C_Sys = 1; C_c = 1;
            end
        else
            C_b = 1; C_c = 1; C_Sys = 1;
        end
    end
else
    if not((state(1) || state(2))) % T1 or T2 On or both
        C_Sys = 1; C_a = 1; C_b = 1; C_c = 1;
    elseif not(state(3) || (state(4))) % A&B off
        C_Sys = 1; C_a = 1; C_b = 1; C_c = 1;
    else
        if state(3)
            C_a = 0;
        else
            C_a = 1; C_Sys = 1;
        end
        if state(4)
            C_b = 0;
            if state(5)
                C_c = 0;
            else
                C_Sys = 1; C_c = 1;
            end
        else
            C_b = 1; C_c = 1; C_Sys = 1;
        end
    end
end
end
```

**Appendix Figure 4- Snapshot: Disconnected Components Function, `Get_Disc.m`**

Number: 1 Author: Simon Tindemans Date: Indeterminate
Simon Tindemans: ok, but does not take into account the effect of feeder faults on the overall load level.

## D.2 Time-Sequential Monte Carlo

```
% Main Script
SAMPLE_SIZE = 1000;
% number of independent years to simulate
DELTA_T = 1/24;
% 1 hour = 1/24 days (using consistent units of time (day) throughout)
RUN_TIME = 365;  % 365 days
% initialise value arrays to zero
cml_values = zeros(1,SAMPLE_SIZE);
ci_values = zeros(1,SAMPLE_SIZE);
EENS_values = zeros(1,SAMPLE_SIZE);
steps_per_run = RUN_TIME / DELTA_T;
%start timer to compute total computation time
for year = 1:SAMPLE_SIZE
    current_state = Get_Sample();
    [Sys,A,B,C] = Is_it_disconnect(current_state);
    current_time = 0;
    for time_index=1:steps_per_run
        % store the previously disconnected customer fraction. This is used for the
CI computation.
            Sys_old = Sys;
            A_old = A;
            B_old = B;
            C_old = C;
            [Sys,A,B,C,C_Load] = Is_it_disconnect(current_state);
            % CML
            cml_values(year) = cml_values(year) + (24*60*DELTA_T*A)*5 ...
                + (24*60*DELTA_T*B)*4 + (24*60*DELTA_T*C)*2;
            % CI
            ci_values(year) = ci_values(year) + max(0, A - A_old)*5 ...
                + max(0, B - B_old)*4 + max(0, C - C_old)*2;
            % EENS
            EENS_values(year) = EENS_values(year) + (A)*C_Load*5 + (B)*C_Load*4 +
(C)*C_Load*2;
            current_state = Next_Time_Step(current_state, DELTA_T);
    end
end
% compute the MC estimate of the LOLP, CML, CI and EENS and their error estimates
cml_estimate = (mean(cml_values))/11;
cml_se = std(cml_values)/(11*sqrt(SAMPLE_SIZE));
LOLP_estimate = cml_estimate/(24*60*365);
LOLP_se = (std(cml_values)/(11*sqrt(SAMPLE_SIZE)))/(24*60*365);
ci_estimate = (mean(ci_values))*(100/11);
ci_se = std(ci_values)/(11*sqrt(SAMPLE_SIZE));
EENS_estimate = (mean(EENS_values))/11;
EENS_se = std(EENS_values)/(11*sqrt(SAMPLE_SIZE));
% Printing
fprintf('\n LOLP  = %.5f - [%.5f, %.5f] (95%% confidence)\n',LOLP_estimate,...
    LOLP_estimate - 1.96*LOLP_se, LOLP_estimate + 1.96*LOLP_se);
fprintf(' CML  = %.5f - [%.5f, %.5f] (95%% confidence)\n',cml_estimate,...
    cml_estimate - 1.96*cml_se, cml_estimate + 1.96*cml_se);
fprintf(' CI  = %.5f - [%.5f, %.5f] (95%% confidence)\n',ci_estimate,...
    ci_estimate - 1.96*ci_se, ci_estimate + 1.96*ci_se);
fprintf(' EENS  = %.5f - [%.5f, %.5f] (95%% confidence)\n\n',EENS_estimate,...
    EENS_estimate - 1.96*EENS_se, EENS_estimate + 1.96*EENS_se);
% Histograms
histogram(cml_values);
xlabel('CML'); ylabel('Years');
figure;
histogram(ci_values);
xlabel('CI'); ylabel('Years');
figure;
histogram(EENS_values);
xlabel('EENS'); ylabel('Years');
```

**Appendix Figure 5- Time-Sequential: Main Script – RUN.m**

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Number: 1 Author: Simon Tindemans Date: Indeterminate
Simon Tindemans: There is a missing factor 100 here!

```
function Load = Load_Profile(Hr, Cust)
LD_Pf = [0.30 0.25 0.23 0.22 0.21 0.25 0.38 0.52 0.53 0.48 0.45 0.46 0.49 0.44 0.43
0.48 0.65 0.83 0.86 0.81 0.75 0.70 0.60 0.43];
MW_LD_Pf = Cust*LD_Pf;
Load = MW_LD_Pf(Hr);
end
```

**Appendix Figure 6- Time-Sequential: Load Profile Function, `Load_Profile.m`**

```
function [ States ] = Get_Sample()
T1 = rand < 0.99972; T2 = rand < 0.99972; F1 = rand < 0.999452;
F2 = rand < 0.999178; F3 = rand < 0.999315; load = Load_Profile(1, 11000);
States = [T1 T2 F1 F2 F3 load 1];
end
```

**Appendix Figure 7- Time-Sequential: Get Samples Function, `Get_Sample.m`**

```
function [ C_Sys, C_a, C_b, C_c, Load] = Is_it_disconnect( state )
    C_Sys = 0; C_a = 0; C_b = 0; C_c = 0;
if (state(6) > 8000)
    if not((state(1) && state(2))) % T1 and T2 Off
        C_Sys = 1; C_a = 1; C_b = 1; C_c = 1;
    elseif not(state(3) || (state(4))) % A&B off
        C_Sys = 1; C_a = 1; C_b = 1; C_c = 1;
    else
        if state(3)
            C_a = 0;
        else
            C_a = 1; C_Sys = 1;
        end
        if state(4)
            C_b = 0;
            if state(5)
                C_c = 0;
            else
                C_Sys = 1; C_c = 1;
            end
        else
            C_Sys = 1; C_c = 1; C_b = 1;
        end
    end
else
    if not((state(1) || state(2)))
        C_Sys = 1; C_a = 1; C_b = 1; C_c = 1;
    elseif not(state(3) || (state(4))) % A&B off
        C_Sys = 1; C_a = 1; C_b = 1; C_c = 1;
    else
        if state(3)
            C_a = 0;
        else
            C_a = 1; C_Sys = 1;
        end
        if state(4)
            C_b = 0;
            if state(5)
                C_c = 0;
            else
                C_Sys = 1; C_c = 1;
            end
        else
            C_b = 1; C_c = 1; C_Sys = 1;
        end
    end
end
Load = state(6);
end
```

**Appendix Figure 8- Time-Sequential: Disconnected Components Function, `Is_it_disconnect.m`**

```
function [ new_state ] = Next_Time_Step( state, dt )
% component order is: T1 T2 F1 F2 F3
rates_fail = [1/36500 1/36500 1/3650 3/7300 3/1825];
rates_repair = [1/10 1/10 1/2 1/2 2.4];
Cust_Total = 11000;
new_state = state;
% Transformer 1
if state(1)==1 % transformer is up
    if rand() < rates_fail(1)*dt
        new_state(1) = 0;
    end
else % transformer is down
    if rand() < rates_repair(1)*dt
        new_state(1) = 1;
    end
end
% Transformer 2
if state(2)==1 % transformer is up
    if rand() < rates_fail(2)*dt
        new_state(2) = 0;
    end
else % transformer is down
    if rand() < rates_repair(2)*dt
        new_state(2) = 1;
    end
end
% Feeder 1
if state(3)==1 % feeder is available
    if rand() < rates_fail(3)*dt
        new_state(3) = 0;
    end
else % feeder is unavailable
    if rand() < rates_repair(3)*dt
        new_state(3) = 1;
    end
end
% Feeder 2
if state(4)==1 % feeder is available
    if rand() < rates_fail(4)*dt
        new_state(4) = 0;
    end
else % feeder is unavailable
    if rand() < rates_repair(4)*dt
        new_state(4) = 1;
    end
end
% Feeder 3
if state(5)==1 % feeder is available
    if rand() < rates_fail(5)*dt
        new_state(5) = 0;
    end
else  % feeder is unavailable
    if rand() < rates_repair(5)*dt
        new_state(5) = 1;
    end
end
% load
new_state(7) = state(7)+1;
if (new_state(7) == 25)
   new_state(7)=1;
end
new_state(6) = Load_Profile(new_state(7), Cust_Total);
end
```

**Appendix Figure 9- Time-Sequential: Stepping Function, `Next_Time_Step.m`**

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